

JEE Mains 2025 Shift 2 — Q6

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An educated guess is not a guess.

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If

$$\int e^x \left(\frac{x \arcsin x}{\sqrt{1-x^2}} + \frac{\arcsin x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2} \right) dx = g(x) + C$$

where C is the constant of integration, then $g(1/2)$ is?

Solution

It is natural to guess that the anti-derivative might take the form $g(x) = e^x q(x) \arcsin x$. We have

$$g'(x) = e^x q(x) \arcsin x + e^x q'(x) \arcsin x + e^x q(x) \frac{1}{\sqrt{1-x^2}}.$$

We want to try and match that last term to the last term in the integrand so we try the function

$$q(x) = \frac{x}{\sqrt{1-x^2}}$$

which has derivative

$$q'(x) = \frac{1}{\sqrt{1-x^2}} + \frac{x^2}{(1-x^2)^{\frac{3}{2}}}.$$

So we can re-write $g'(x)$ now as

$$\begin{aligned} g'(x) &= e^x \left(\frac{x \arcsin x}{\sqrt{1-x^2}} + \frac{\arcsin x}{\sqrt{1-x^2}} + \frac{x^2 \arcsin x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2} \right) \\ &= e^x \left(\frac{x \arcsin x}{\sqrt{1-x^2}} + \frac{(1-x^2) \arcsin(x) + x^2 \arcsin x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2} \right) \\ &= e^x \left(\frac{x \arcsin x}{\sqrt{1-x^2}} + \frac{\arcsin x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2} \right) \end{aligned}$$

which is the given integrand. So we know that our choice of $g(x)$ does in fact satisfy the given equation. Finally we can just compute

$$g\left(\frac{1}{2}\right) = \frac{\pi}{6} \sqrt{\frac{e}{3}}. \blacksquare$$

Comments: Interestingly enough ChatGPT with thinking mode, struggled significantly to solve this one even after I gave hints. The real trap is that after you compute $g'(x)$ it doesn't look like it matches the integrand, but if you combine like terms correctly, you'll see that it actually does match the integrand.